
Quadrotor Trajectory Tracking Control

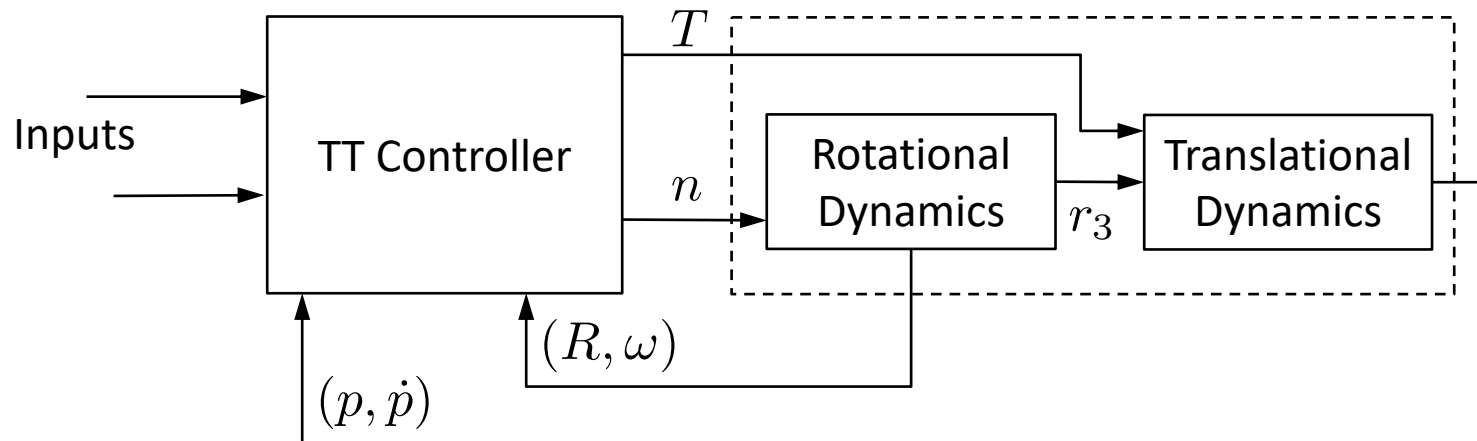
An Introduction

UAVs

MEAer - Autumn Semestre – 2022/2023

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Quadrotor control: **topics for today**



- Define **control objectives**: Trajectory Tracking
- Define the **control strategy**:
 - Hierarchical or Cascade block diagram
 - Outer/Inner loops
- Linear control techniques:
 - **SISO**, specs and design tools
- or **State feedback**

Quadrotor dynamic model

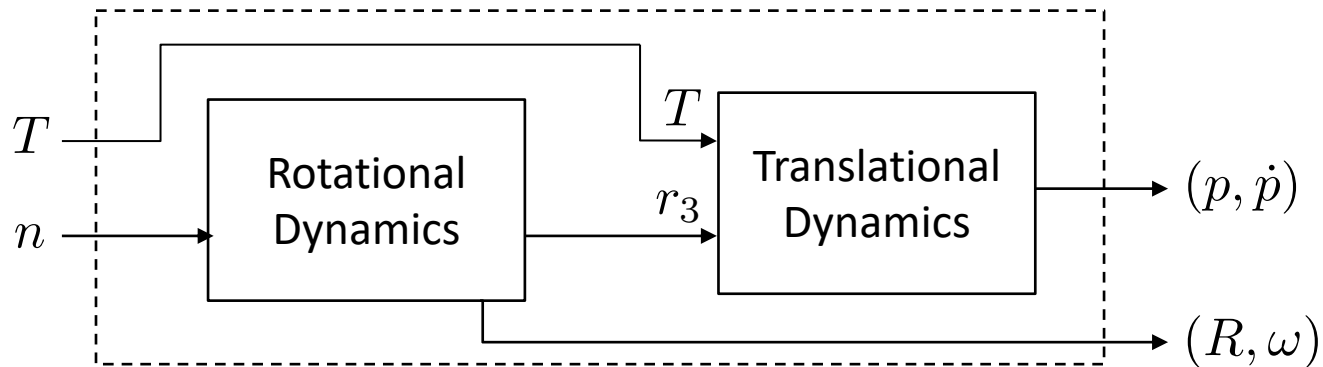
- Equations of motion

$$m\ddot{p} = -Tr_3 + mge_3, \quad r_3 = Re_3$$

$$\dot{R} = RS(\omega)$$

$$J\dot{\omega} = -S(\omega)J\omega + n$$

- Block diagram



Quadrotor dynamic model

- Equations of motion

$$m\ddot{p} = -Tr_3 + mge_3, \quad r_3 = Re_3$$

$$\dot{R} = RS(\omega)$$

$$J\dot{\omega} = -S(\omega)J\omega + n$$

- Exercise:

- Show that for small angles near hover:

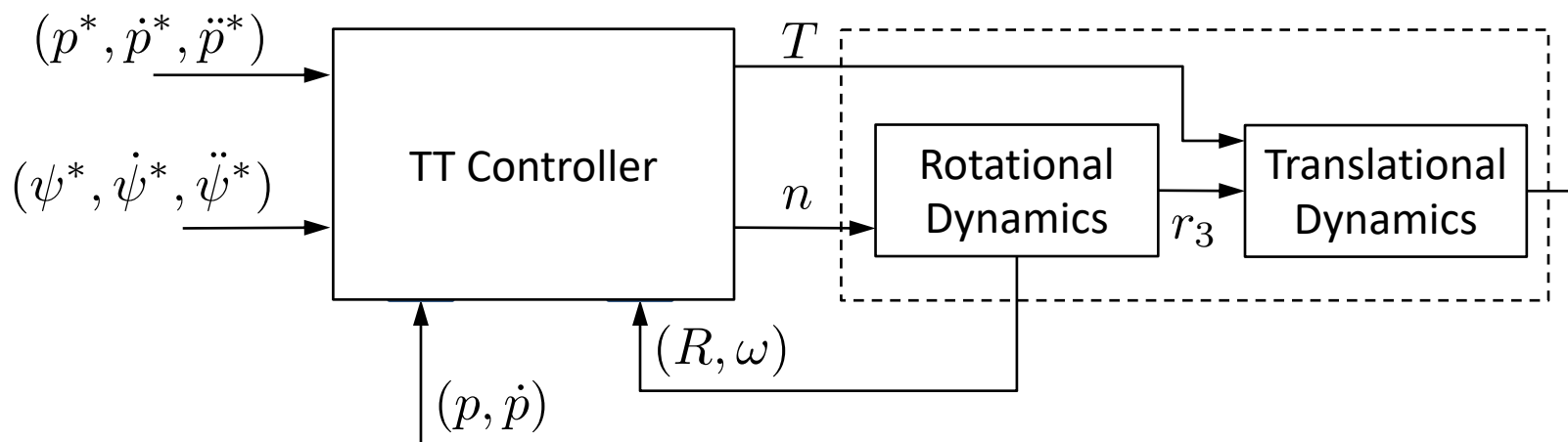
$$\ddot{p} \approx \begin{bmatrix} -g\theta \\ g\phi \\ 0 \end{bmatrix}$$

- Compare with the block diagram

Feedback control for trajectory tracking

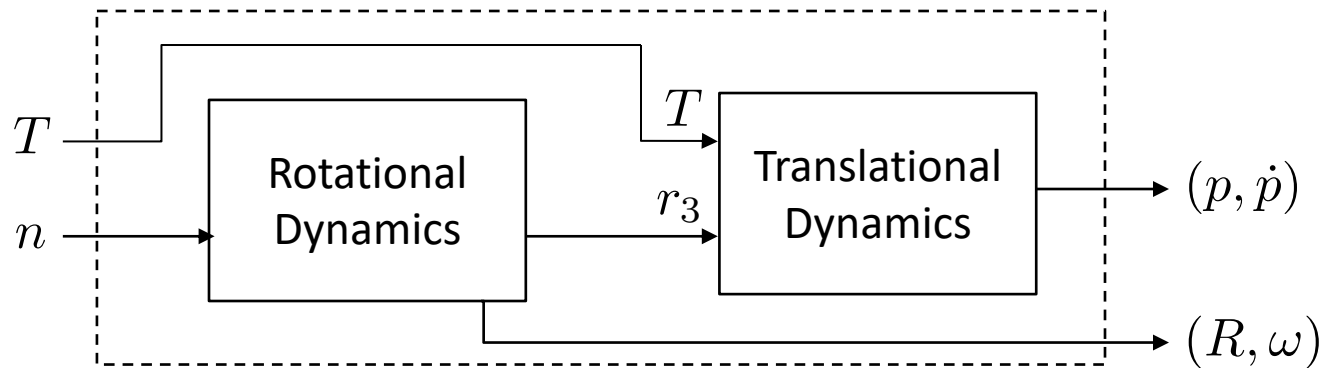
- Control objective:
 - Consider a desired trajectory for the flat output $(p^*(t), \psi^*(t))$
 - Define a control law for (T, n) such that

$$(p(t), \psi(t)) \rightarrow (p^*(t), \psi^*(t)) \text{ as } t \rightarrow +\infty$$



Hierarchical or Cascade control

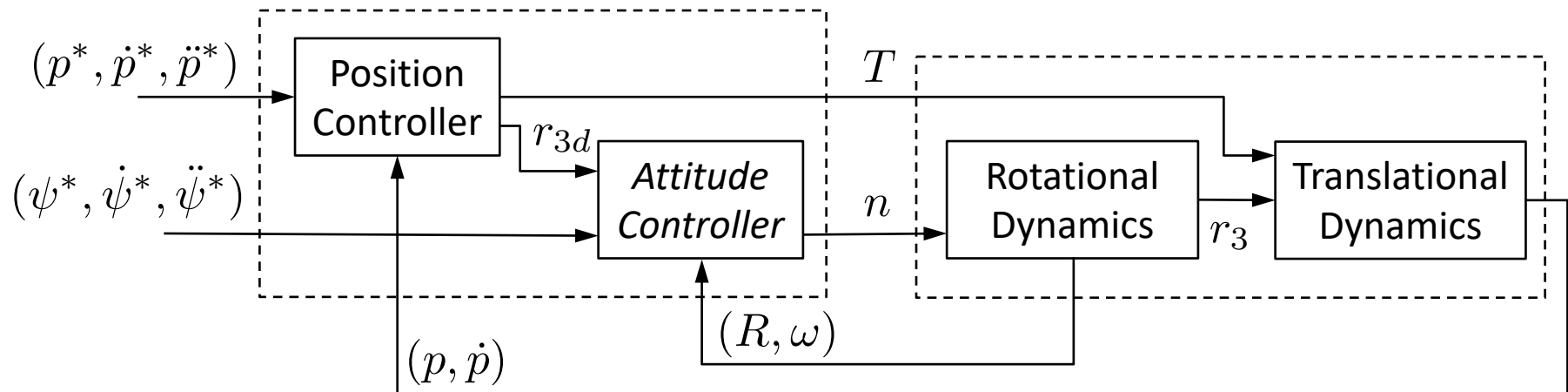
- Block dependencies suggest a hierarchical control structure



- Outer-loop controller for translational dynamics determines the thrust T and desired orientation R_d
 - Goal: track desired position
- Inner-loop controller for rotational dynamics determines the torque n
 - Goal: track desired attitude

Hierarchical or Cascade control

- Inner-outer loop control structure



- Outer-loop controller for translational dynamics determines the thrust T and desired orientation R_d
 - Goal: track desired position
- *Inner-loop controller for rotational dynamics determines the torque n*
 - *Goal: track desired attitude*

Hierarchical or Cascade control

1. Neglect rotational dynamics

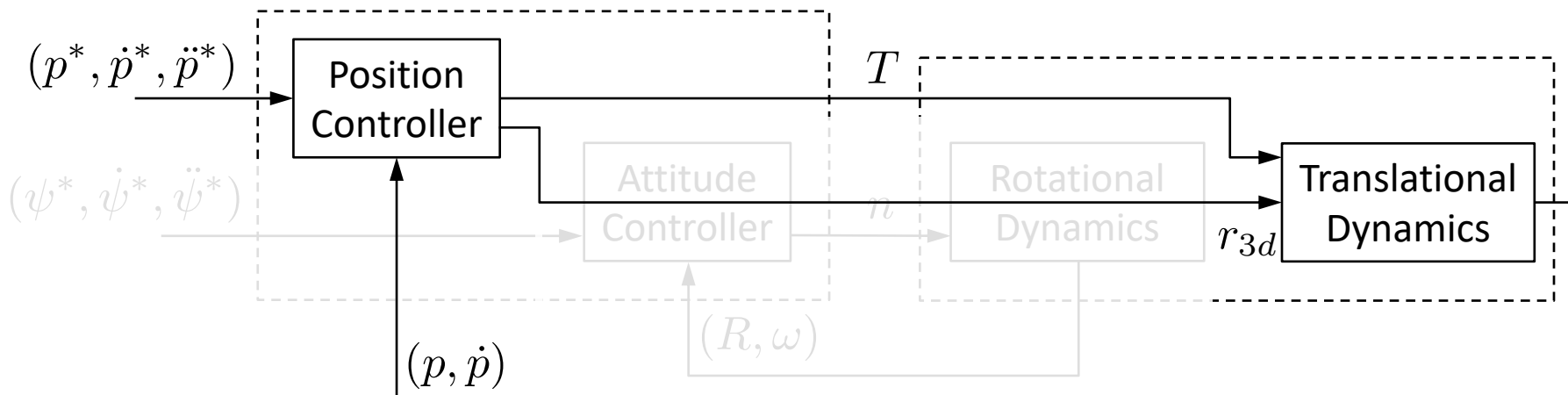
- Replace r_3 by a desired value r_{3d}
- Define a new virtual input for the translational dynamics
 T gives magnitude, r_{3d} gives direction

$$u_T = -\frac{T}{m}r_{3d}$$

$$T = m\|u_T\|, \quad r_{3d} = -\frac{u_T}{\|u_T\|}$$

2. Define a control law for u_T , assuming

$$\ddot{p} = u_T + g \quad u_T = ?$$



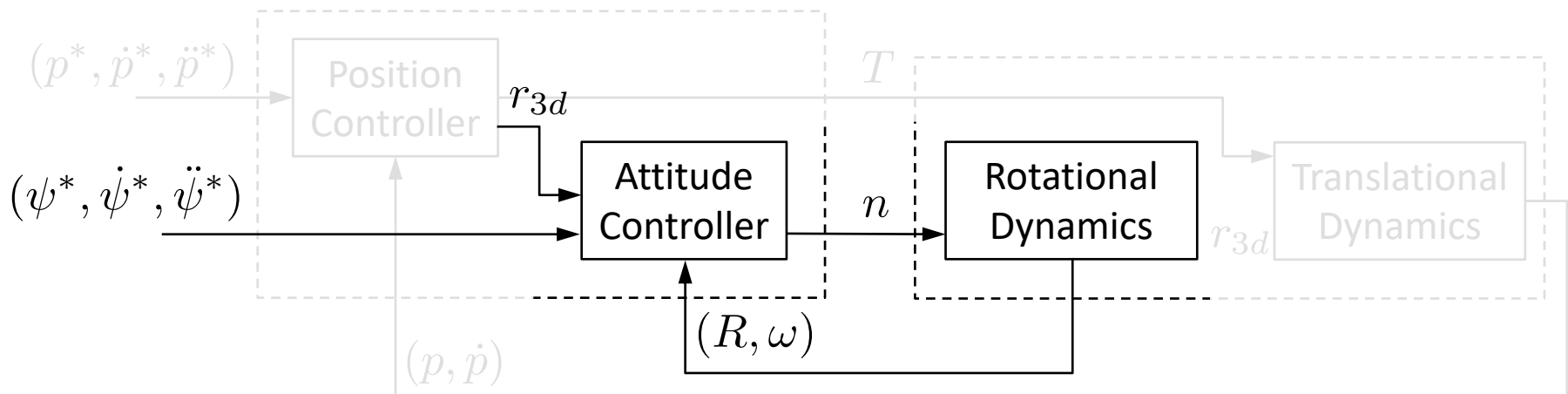
Hierarchical or Cascade control

3. Combine r_{3d} with ψ^* to define a desired attitude R_d

$$R_d = f(r_{3d}, \psi^*) =? \quad \text{or} \quad \begin{aligned} \lambda_d &= (\phi_d, \theta_d, \psi^*) \\ (\phi_d, \theta_d) &= g(r_{3d}, \psi^*) =? \end{aligned}$$

4. Define a control law for n to track the desired attitude R_d or λ_d

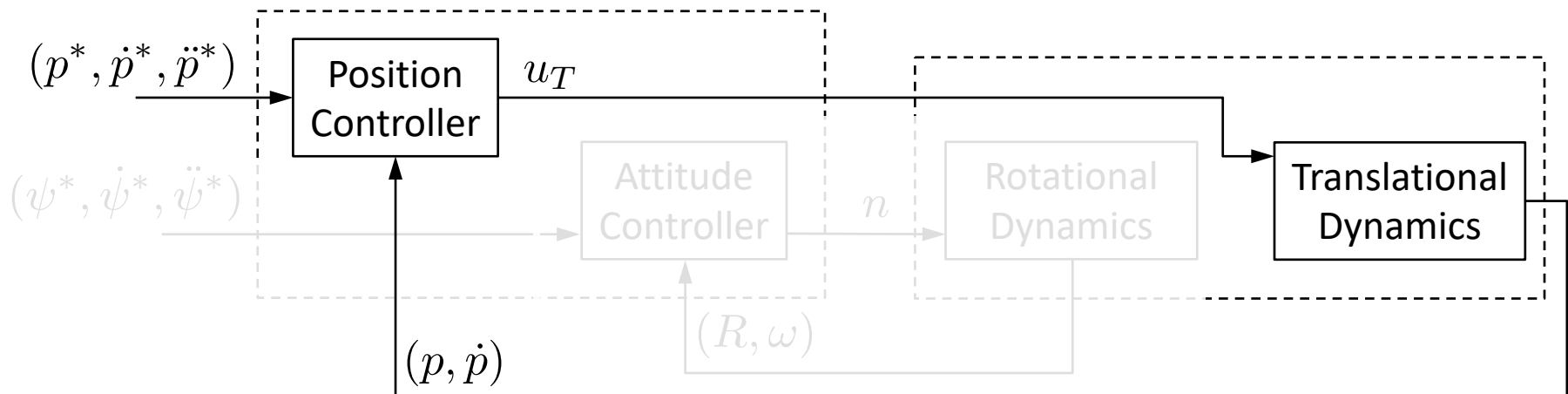
$$\begin{aligned} \dot{R} &= RS(\omega) & \text{or} & & \dot{\lambda} &= Q(\phi, \theta)\omega \\ J\dot{\omega} &= -S(\omega)J\omega + n & & & J\dot{\omega} &= -S(\omega)J\omega + n \end{aligned} \quad n = ?$$



Trajectory Tracking Controller (position)

2. Define a control law for u_T to track a desired trajectory

- Model $\ddot{p} = u_T + g$
- Reference to be tracked $(p^*(t), \dot{p}^*(t), \ddot{p}^*(t))$
- Control law $u_T + g = ?$



Trajectory Tracking Controller (position)

2. Define a control law for u_T to track a desired trajectory

– Model

$$\ddot{p} = u_T + g$$

– Reference to be tracked

$$(p^*(t), \dot{p}^*(t), \ddot{p}^*(t))$$

– Error system

$$e(t) = p(t) - p^*(t), \quad \dot{e}(t) = \dot{p}(t) - \dot{p}^*(t)$$

$$\ddot{e}(t) = u_T(t) + g - \ddot{p}^*(t)$$

– Control law

$$u_T(t) = -k_P e(t) - k_D \dot{e}(t) - g + \ddot{p}^*(t)$$

– Closed-loop error system

$$\ddot{e}(t) = -k_P e(t) - k_D \dot{e}(t)$$

Trajectory Tracking Controller (position)

2. Define a control law for u_T to track a desired trajectory
 - Closed-loop error system

$$\ddot{e}(t) = -k_P e(t) - k_D \dot{e}(t)$$

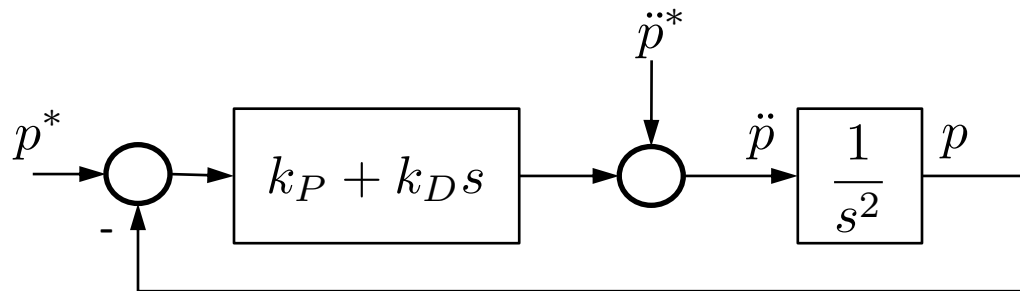
- Stability

$$k_P > 0, k_D > 0 \quad \longrightarrow \quad \text{Asymptotic Stability}$$

- Performance Specifications
 - Settling time
 - Overshoot
 - Disturbance rejection
 - Noise attenuation
 - Gain and phase margin

Trajectory Tracking Controller (position)

- Classical SISO system analysis



- Closed-loop system in time domain

$$\ddot{p} = \underbrace{-k_P(p - p^*) - k_D(\dot{p} - \dot{p}^*)}_{\text{feedback}} + \underbrace{\ddot{p}^*}_{\text{feedforward}}$$

- Closed-loop transfer function

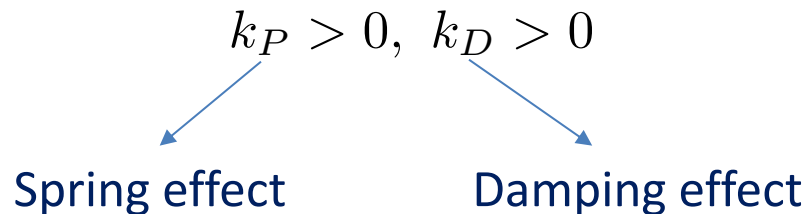
$$\frac{P(s)}{P^*(s)} = \frac{k_P + k_D s}{s^2 + k_D s + k_P} \quad \text{compare with} \quad G(s) = \frac{\omega_n^2}{s^2 + 2\xi\omega_n s + \omega_n^2}$$

Classical SISO system analysis

- Closed-loop transfer function

$$\frac{P(s)}{P^*(s)} = \frac{k_P + k_D s}{s^2 + k_D s + k_P} \quad \text{compare with} \quad G(s) = \frac{\omega_n^2}{s^2 + 2\xi\omega_n s + \omega_n^2}$$

- Stability: poles with negative real part



- Decrease overshoot (increase damping coefficient)

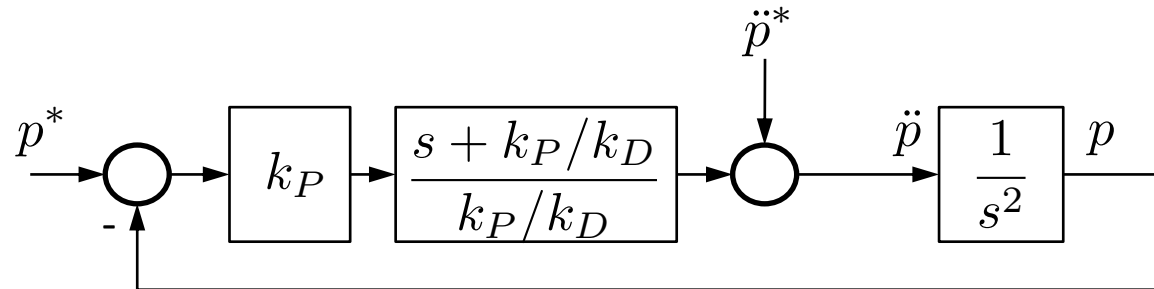
$$\xi = k_D / (2\sqrt{k_P}) \qquad k_D \uparrow \Rightarrow \xi \uparrow$$

$$(k_D \geq 2\sqrt{k_P}) \Rightarrow (\xi \geq 1) \qquad k_P \uparrow \Rightarrow \xi \downarrow$$

- Decrease settling time $k_D \uparrow \Rightarrow \xi\omega_n \uparrow$

Classical SISO system analysis

- Root-locus



- Closed-loop transfer function

$$\frac{P}{P^*} = \frac{k \frac{s+z}{z s^2}}{1 + k \frac{s+z}{z s^2}}$$

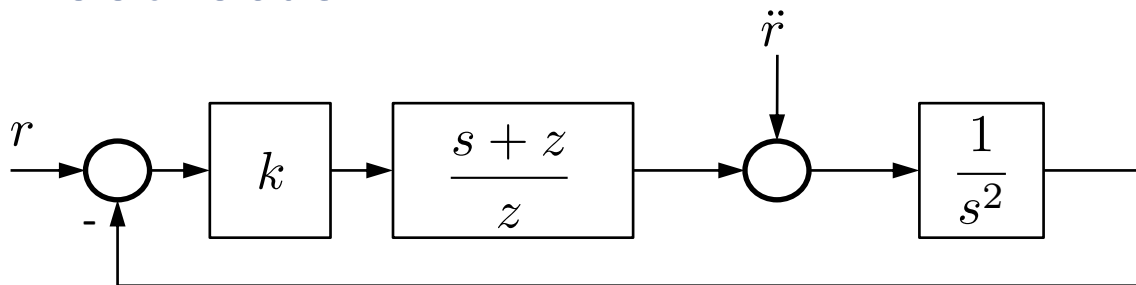
$$z = \frac{k_P}{k_D}, \quad k = k_P$$

- Open-loop transfer function $G(s) = k \frac{s+z}{z s^2}$

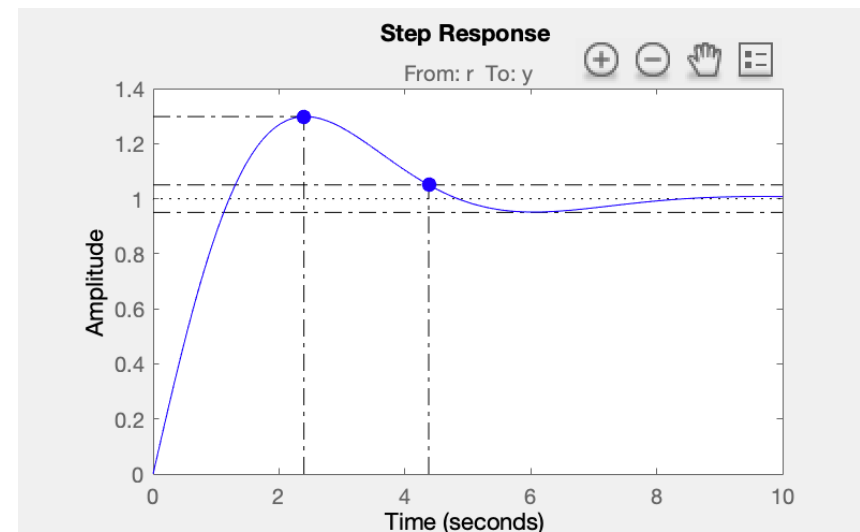
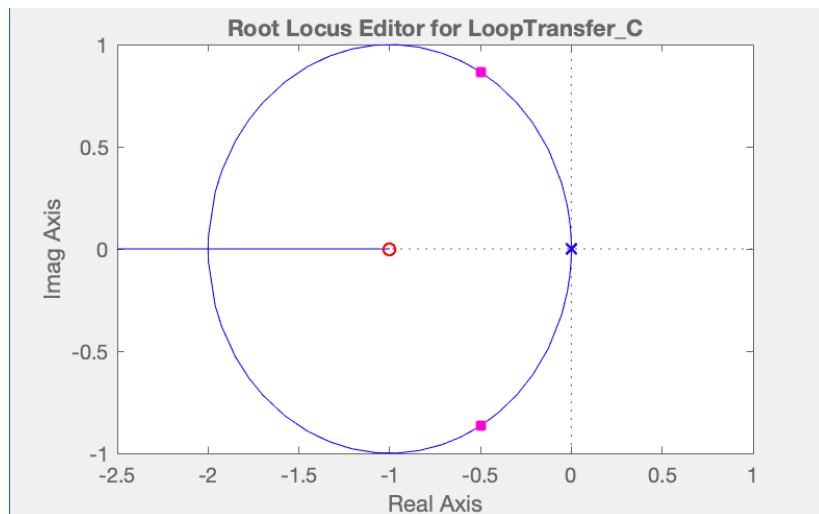
- Characteristic equation $1 + k \frac{s+z}{z s^2} = 0$

Classical SISO system analysis

- Root-locus

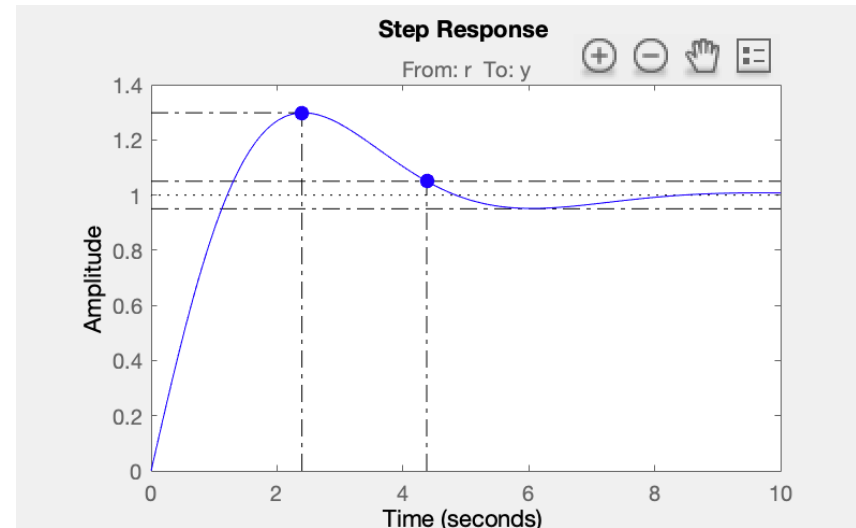
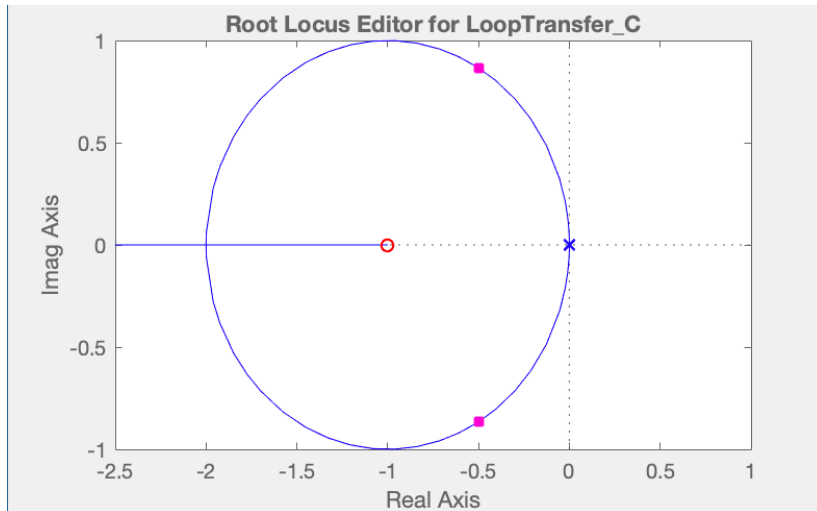


$$k = k_P = 1, \quad z = \frac{k_P}{k_D} = 1$$

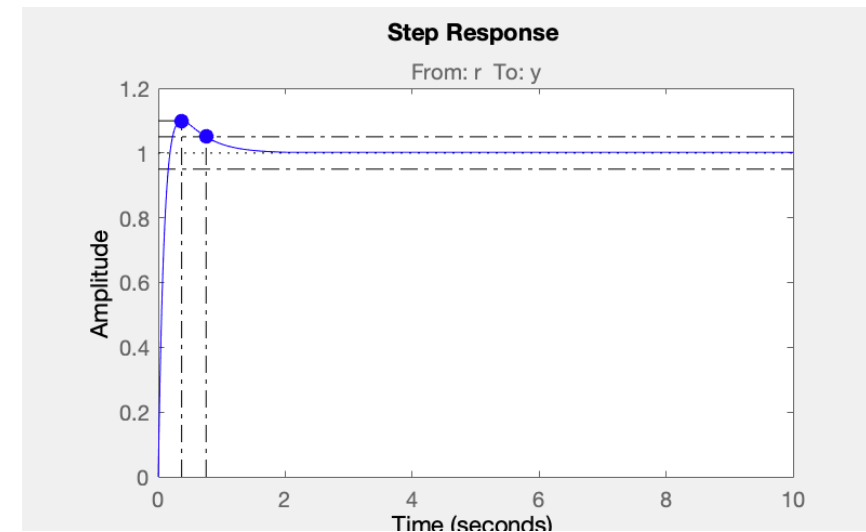
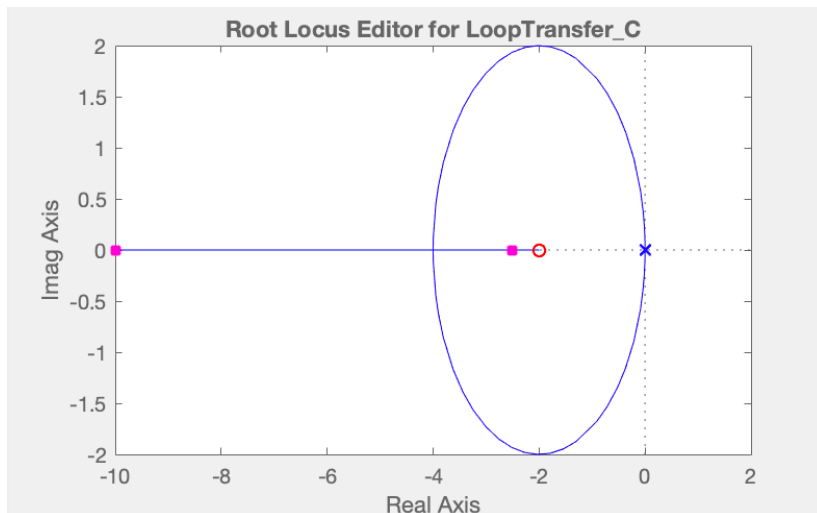


Control design based on Root-locus

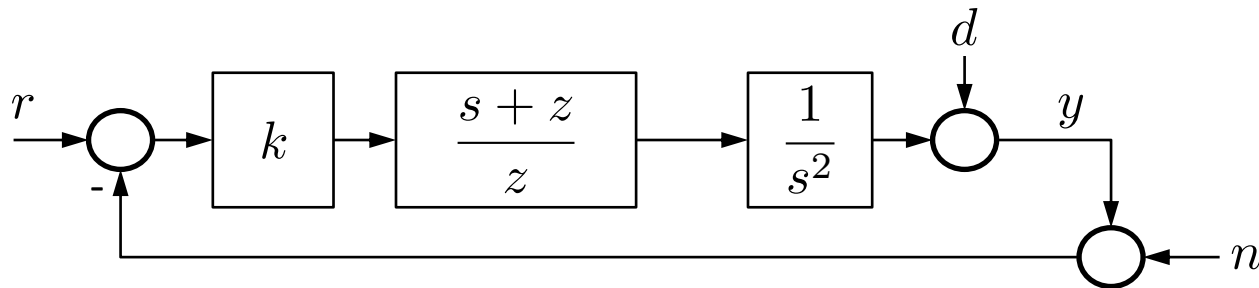
$$k = 1, \quad z = 1$$



$$k = 24, \quad z = 2$$



Control design based on Loop-Shaping



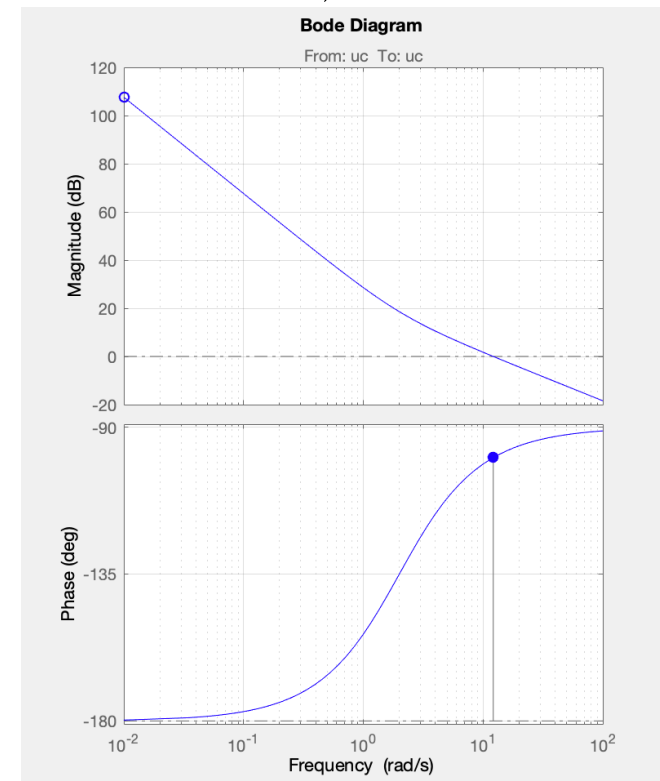
- Bode plot for open-loop transfer function

- $G_M = ?$
- $P_M = ?$
- Noise and disturbance attenuation

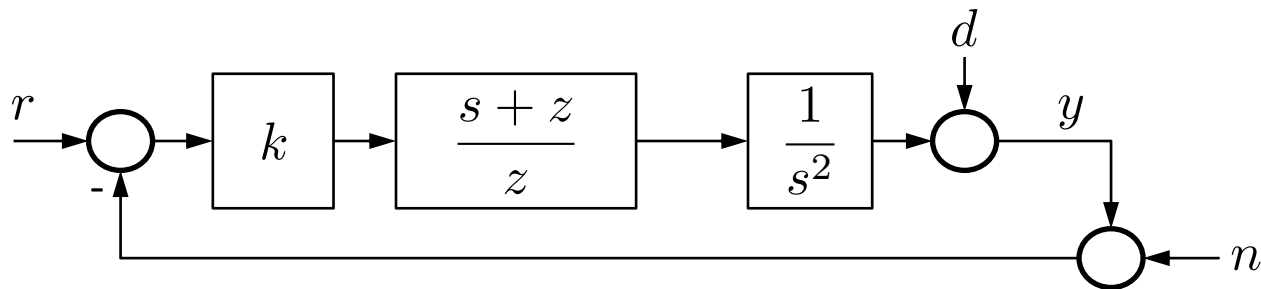
$$\left| \frac{Y(j\omega_n)}{N(j\omega_n)} \right| \leq ?, \omega_n \in [\bar{\omega}_n, \underline{\omega}_n]$$

$$\left| \frac{Y(j\omega_d)}{D(j\omega_d)} \right| \leq ?, \omega_d \in [\bar{\omega}_d, \underline{\omega}_d]$$

$$k = 24, \quad z = 2$$



Control design based on Loop-Shaping



– $G_M = ?$

$$G_M = +\infty$$

– $P_M = ?$

$$P_M = 80.7^\circ$$

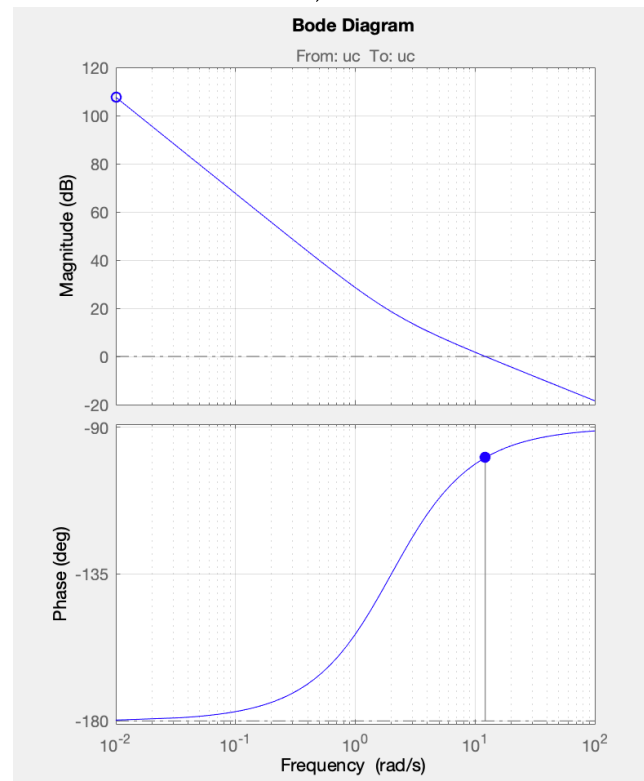
$$\omega_c = 12.2 \text{ rad s}^{-1}$$

– Maximum tolerable time delay

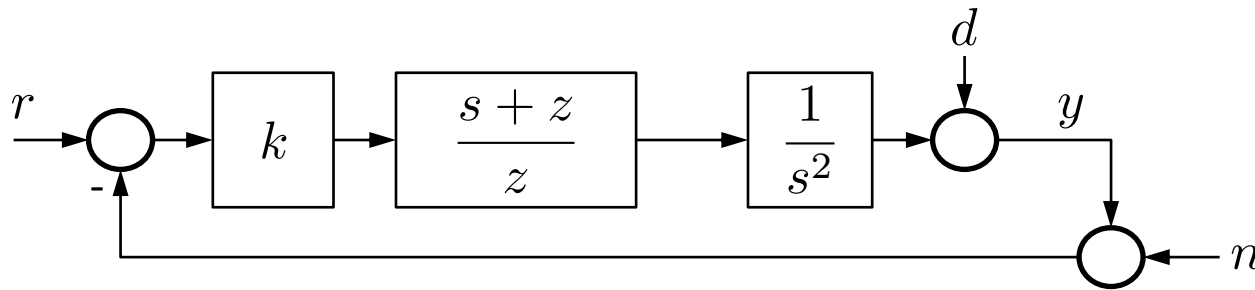
$$\tau \omega_c < \tau_c \omega_c = P_M$$

$$\tau < \tau_c = 0.116 \text{ s}$$

$$k = 24, \quad z = 2$$



Control design based on Loop-Shaping



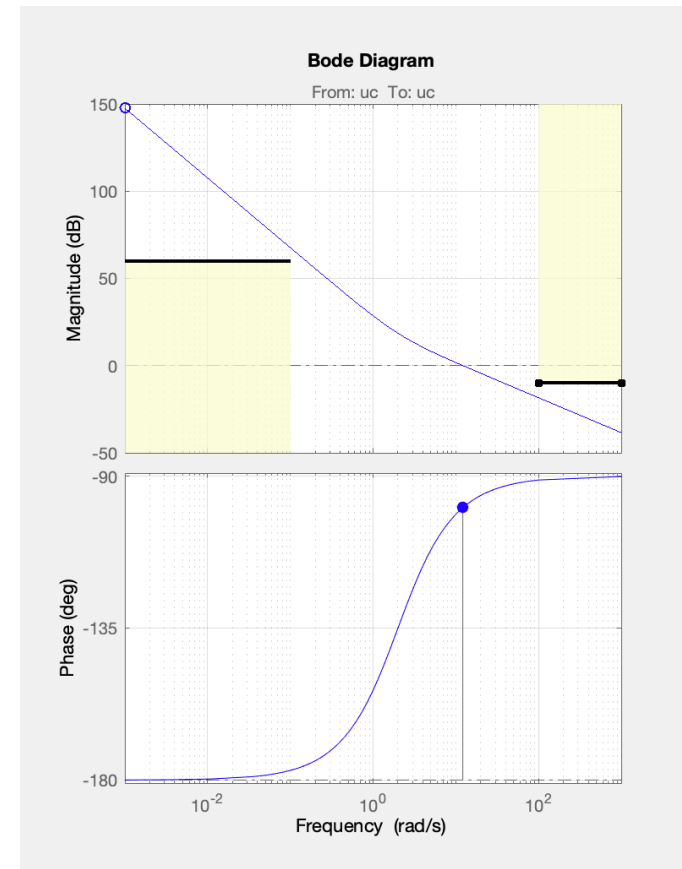
- Noise reduction

$$\left| \frac{Y(j\omega_n)}{N(j\omega_n)} \right|_{dB} \leq -10 \text{ dB}$$

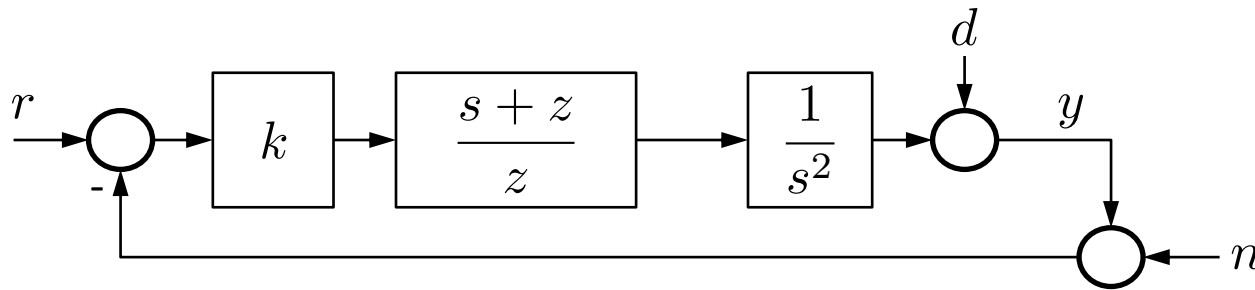


$$|G(j\omega_n)|_{dB} \leq -10 \text{ dB}$$

$$\omega_n \in [10^2, 10^3]$$



Control design based on Loop-Shaping



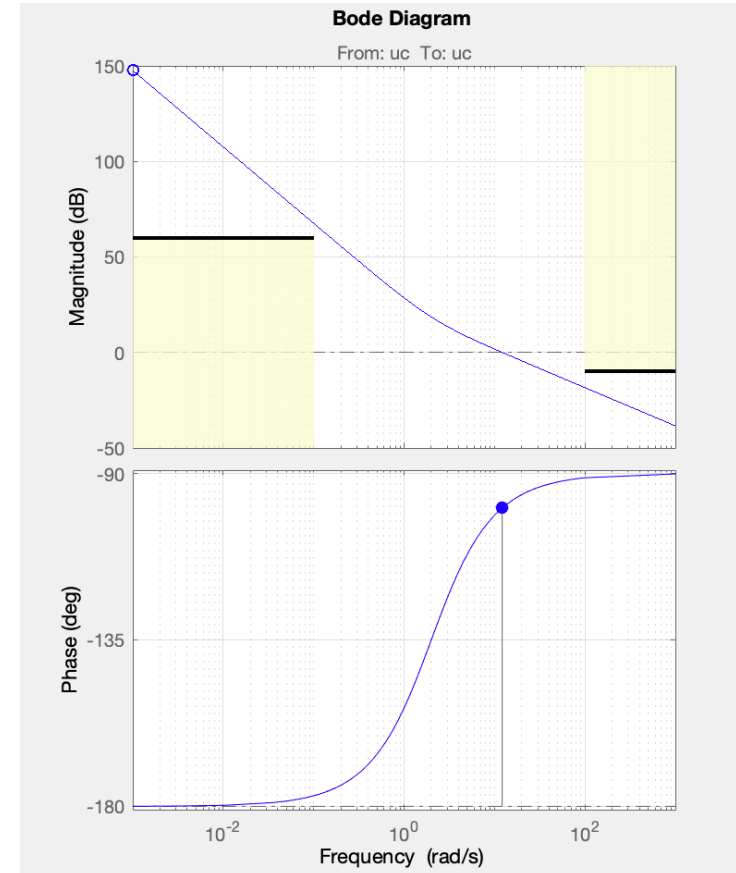
- Disturbance attenuation

$$\left| \frac{Y(j\omega_d)}{D(j\omega_d)} \right|_{dB} \leq -80 \text{ dB}$$

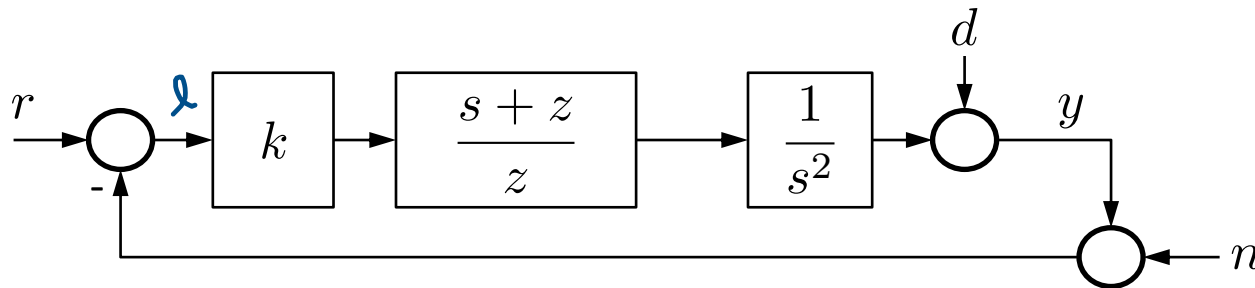


$$|G(j\omega_d)|_{dB} \geq 80 \text{ dB}$$

$$\omega_d \in [10^{-2}, 10^{-1}]$$



Control design based on Loop-Shaping



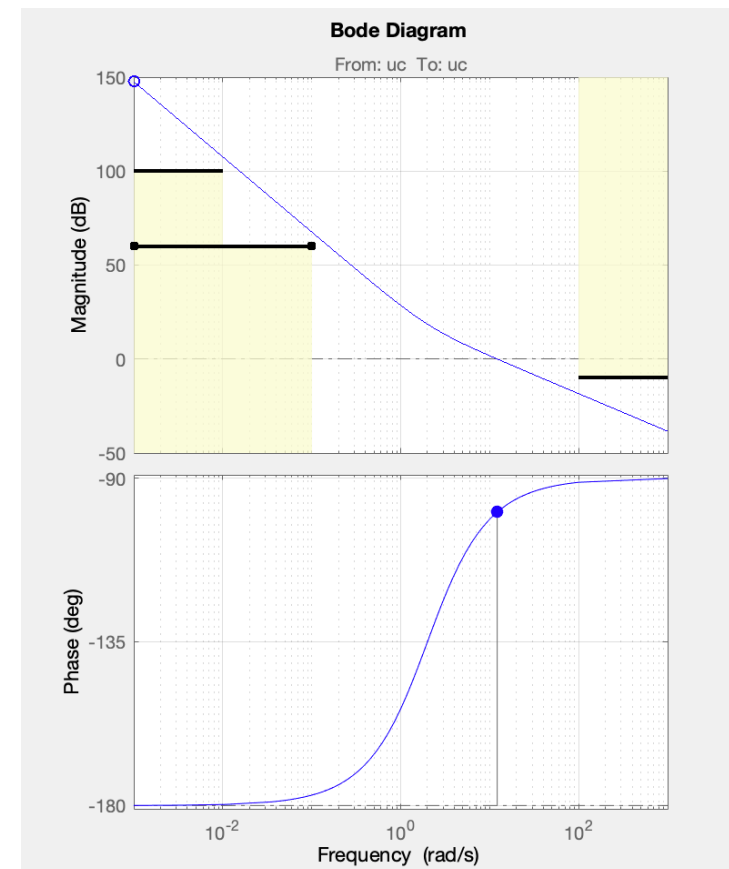
- Reference Following

$$\left| \frac{E(j\omega_d)}{R(j\omega_d)} \right|_{dB} \leq -100 \text{ dB}$$

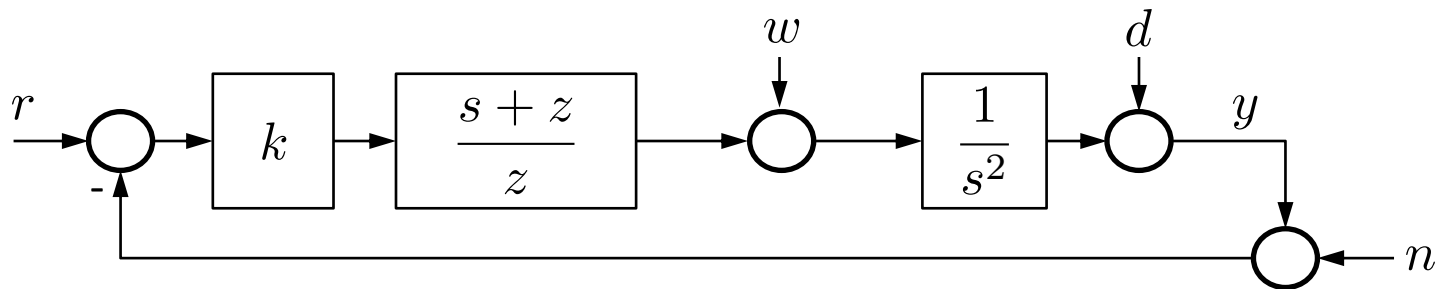


$$\underline{|G(j\omega_r)|_{dB} \geq 100 \text{ dB}}$$

$$\omega_r \in [0, 10^{-2}]$$



Disturbance Rejection



- Wind disturbance rejection


 wind disturbance as acceleration input

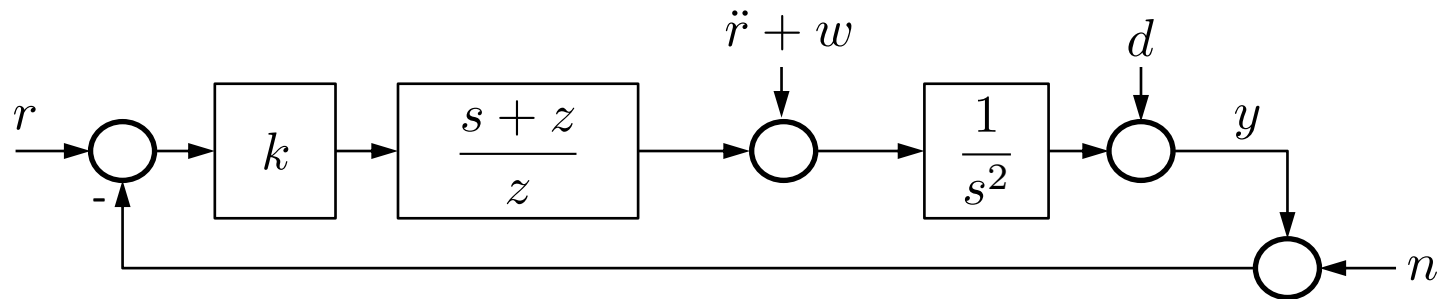
$$Y(s) = D(s) + \frac{1}{s^2} [W(s) + k(s+z)(R(s) - Y(s) - N(s))]$$

$$Y(s) = \frac{s^2}{s^2 + C(s)} D(s) + \frac{1}{s^2 + C(s)} W(s) + \frac{C(s)}{s^2 + C(s)} (R(s) - N(s))$$

$$\lim_{t \rightarrow \infty} y(t) = \lim_{s \rightarrow 0} sY(s)$$

- Does the system reject a constant wind disturbance ?

Disturbance Rejection



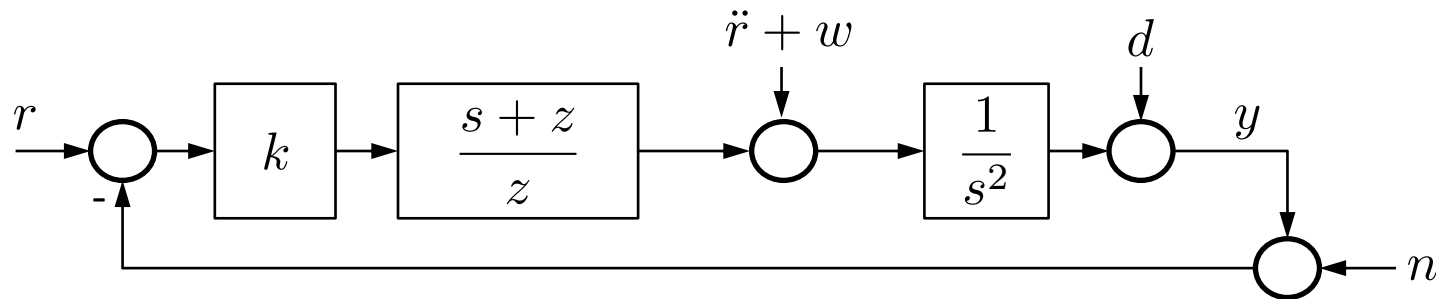
- Wind disturbance rejection
 - Does the system reject a constant wind disturbance?
 - No.

$$\frac{Y(s)}{W(s)} = \frac{1}{s^2 + C(s)}, \quad W(s) = w_0/s$$

- Application of the Final Value Theorem

$$\lim_{t \rightarrow \infty} y(t) = \lim_{s \rightarrow 0} sY(s) = \frac{w_0}{C(0)} = \frac{w_0}{k}$$

Disturbance Rejection



- Wind disturbance rejection
 - Does the system reject a constant wind disturbance? No.
 - Also easy to see in the time domain

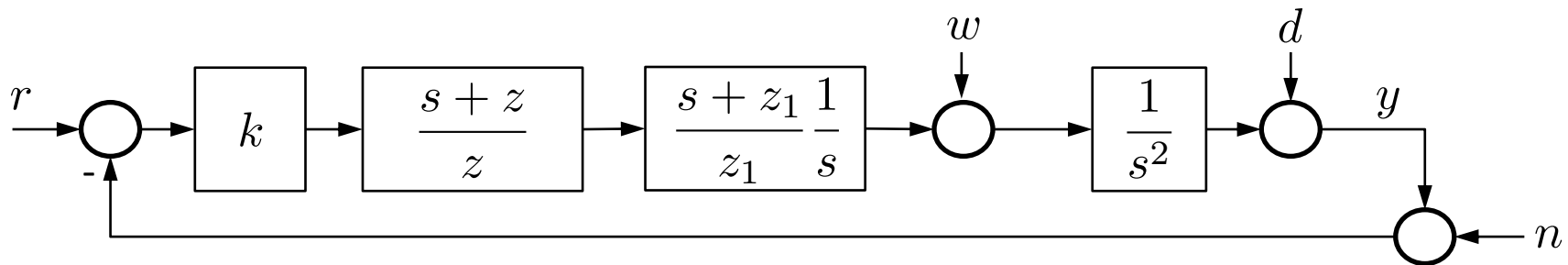
$$\ddot{y} + \frac{k}{z}(\dot{y} - \dot{r}) + k(y - r) = (\ddot{r} + w) + \ddot{d}$$

$$\ddot{e} + \frac{k}{z}\dot{e} + ke = w + \ddot{d}$$

- Stable equilibrium point

$$e^* = \frac{w_0}{k} \Leftrightarrow y^*(t) = r(t) + \frac{w_0}{k}$$

Disturbance Rejection – Integral Action



- Wind disturbance rejection
 - Solution: add integral action -> PID controller

$$C(s) = k_P + \frac{k_I}{s} + k_D s = \frac{k_D s^2 + k_P s + k_I}{s}$$

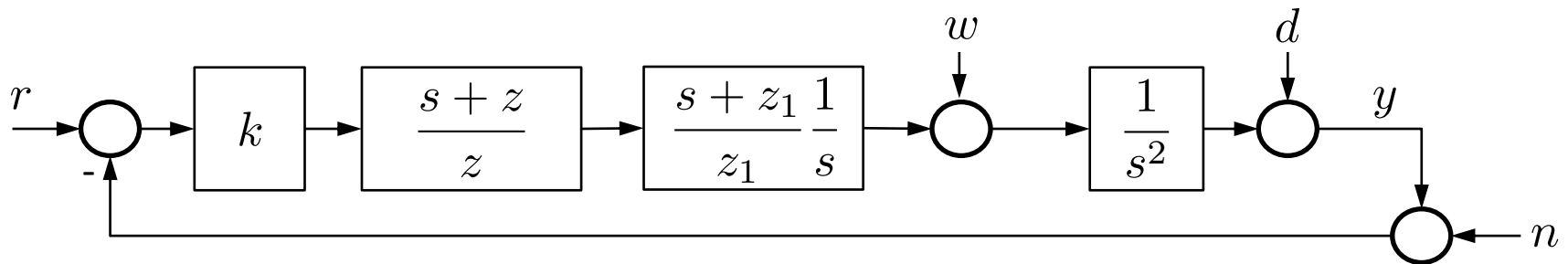
$$\begin{aligned} \frac{Y(s)}{W(s)} &= \frac{1}{s^2 + C(s)} \\ &= \frac{s}{s^2 + k_D s^2 + k_P s + k_I} \end{aligned}$$

$$W(s) = \frac{w_0}{s}, \quad \lim_{t \rightarrow \infty} y(t) = \lim_{s \rightarrow 0} sY(s) = 0$$

Experimental results



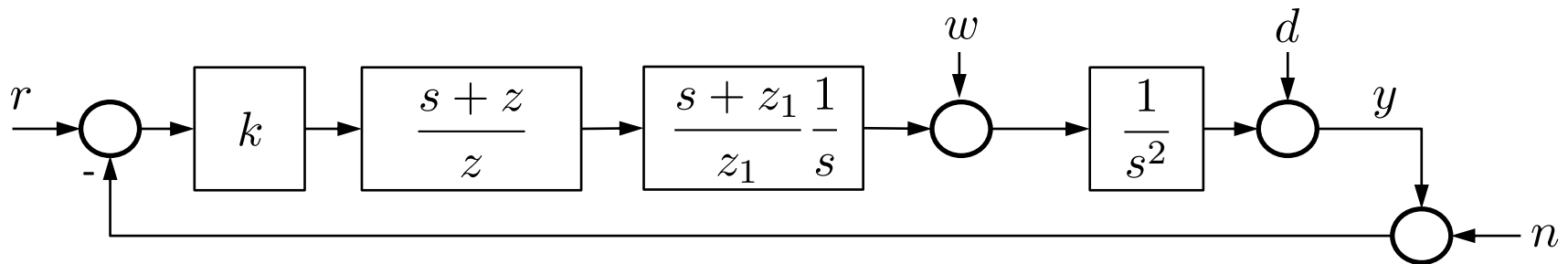
Disturbance Rejection – Integral Action



- Strategy to go from PD to PID controller

$$\begin{array}{ccc}
 C_{PD}(s) = k_P + k_D s & & C_{PID}(s) = k_P + \frac{k_I}{s} + k_D s \\
 \downarrow & \text{Add zero and} & \uparrow \\
 C_{PD}(s) = k \frac{s+z}{z} & \xrightarrow{\text{integrator}} & C_{PID}(s) = k \frac{(s+z)(s+z_1)}{z z_1 s}
 \end{array}$$

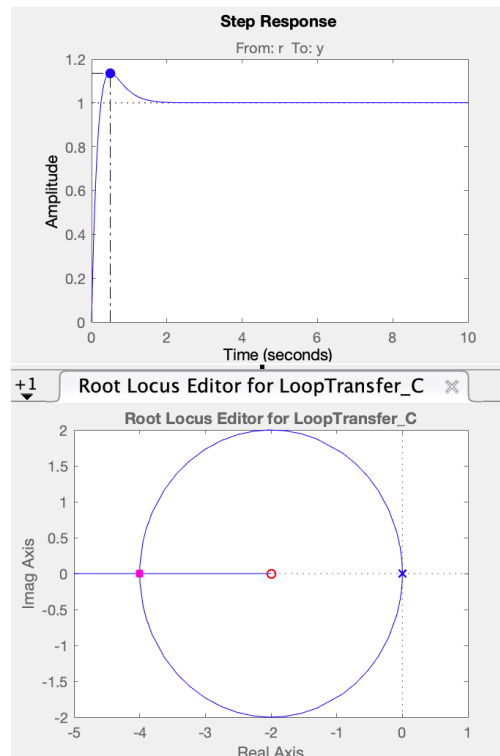
Disturbance Rejection – Integral Action



– keep zero at $-z$, add integrator and zero at $-z_1$

$$k = 16$$

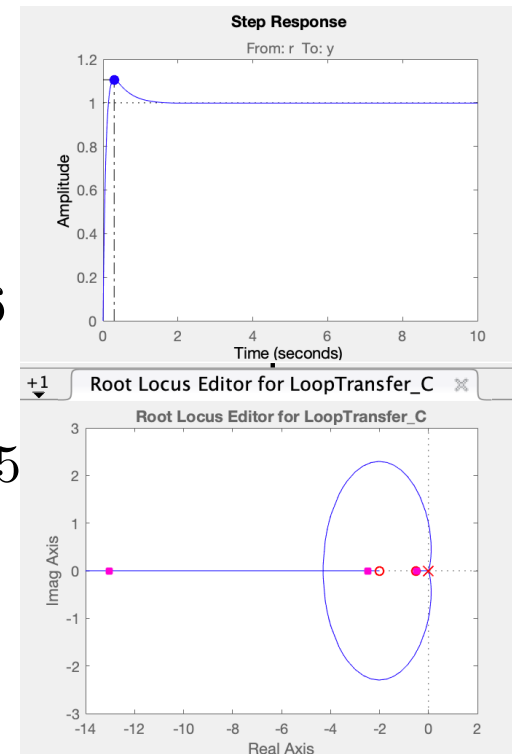
$$z = 2$$



$$k = 16$$

$$z = 2$$

$$z_1 = 0.5$$



Disturbance Rejection: **exercise**

- Wind disturbance rejection
 - Repeat the analysis if the wind disturbance is a velocity input

Hierarchical control

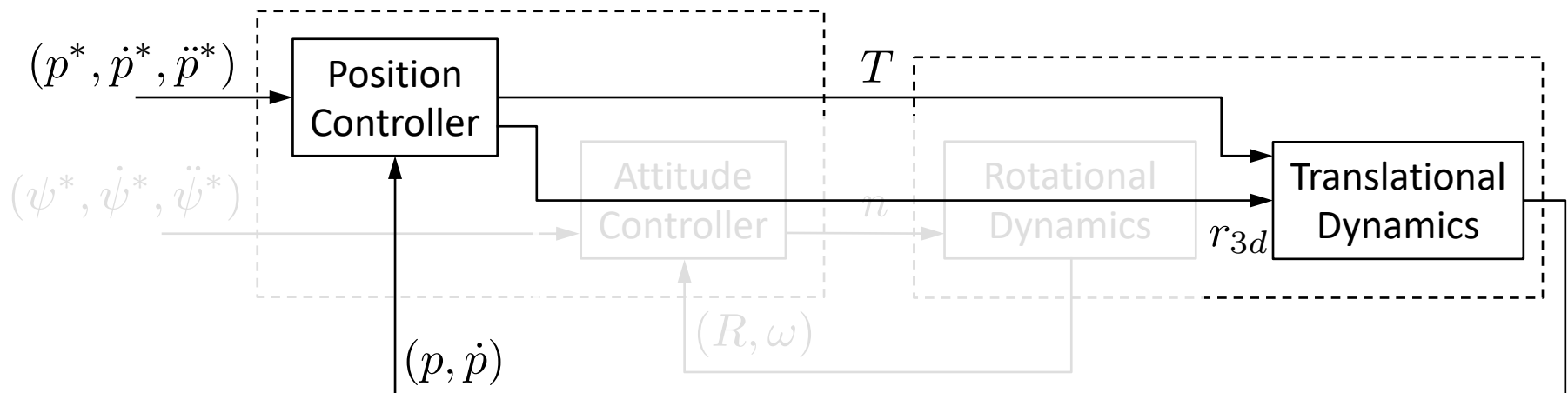
1. Neglect rotational dynamics

- Replace r_3 by a desired value r_{3d}
- Define a new virtual input for the translational dynamics $u_T = -\frac{T}{m}r_{3d}$
 T gives magnitude, r_{3d} gives direction

$$T = m\|u_T\|, \quad r_{3d} = -\frac{u_T}{\|u_T\|}$$

2. Define a control law for u_T , assuming $\ddot{p} = u_T + ge_3$

$$u_T(t) = -k_P(p - p^*) - k_D(\dot{p} - \dot{p}^*) - k_I \int (p - p^*) d\tau - ge_3 + \ddot{p}^*(t)$$



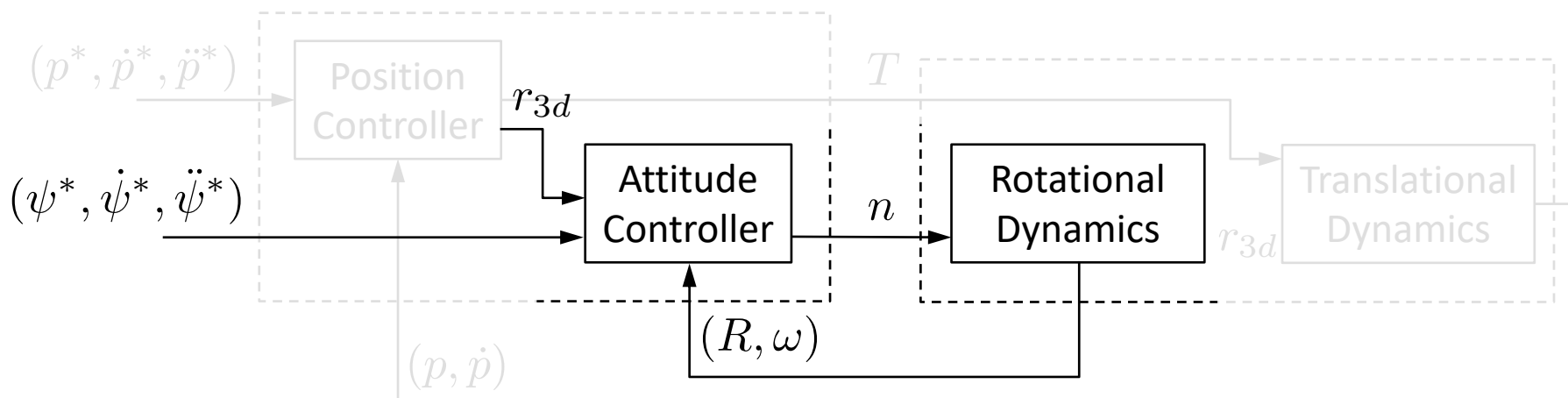
Hierarchical control

3. Combine r_{3d} with ψ^* to define a desired attitude R_d

$$R_d = f(r_{3d}, \psi^*) =? \quad \text{or} \quad \begin{aligned} \lambda_d &= (\phi_d, \theta_d, \psi^*) \\ (\phi_d, \theta_d) &= g(r_{3d}, \psi^*) =? \end{aligned}$$

4. Define a control law for n to track the desired attitude R_d or λ_d

$$\begin{aligned} \dot{R} &= RS(\omega) \\ J\dot{\omega} &= -S(\omega)J\omega + n \end{aligned} \quad \text{or} \quad \begin{aligned} \dot{\lambda} &= Q(\phi, \theta)\omega \\ J\dot{\omega} &= -S(\omega)J\omega + n \end{aligned} \quad n = ?$$



Hierarchical control

3. Combine r_{3d} with ψ^* to define a desired attitude R_d

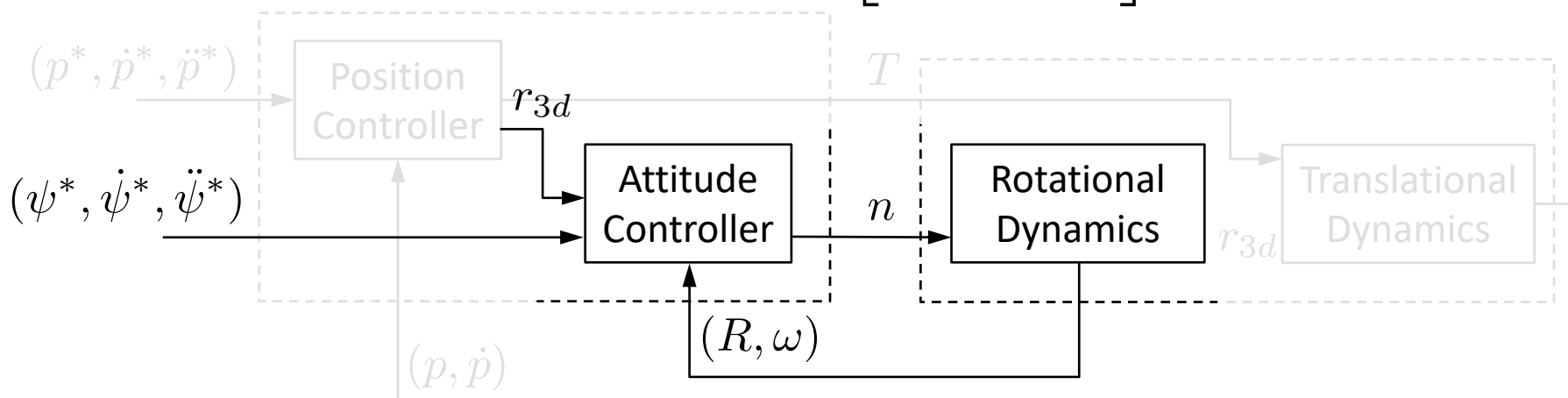
$$R_d = f(r_{3d}, \psi^*) =? \quad \text{or} \quad \lambda_d = (\phi_d, \theta_d, \psi^*)$$

$$(\phi_d, \theta_d) = g(r_{3d}, \psi^*) =?$$

- References for the attitude controller $\lambda_d = (\phi_d, \theta_d, \psi^*)$

$$r_{3d} = R_z(\psi^*)R_y(\theta_d)R_x(\phi_d)e_3$$

$$\begin{bmatrix} \cos \phi_d \sin \theta_d \\ -\sin \phi_d \\ \cos \phi_d \cos \theta_d \end{bmatrix} = R_z(-\psi^*)r_{3d}$$



Attitude Controller

- Linearize model about hover condition $\lambda_0 = [0 \ 0 \ \psi^*]^T$

$$\dot{\lambda} = Q(\phi, \theta)\omega$$

$$\omega_0 = [0 \ 0 \ 0]^T$$

$$J\dot{\omega} = -S(\omega)J\omega + n$$

$$n_0 = [0 \ 0 \ 0]^T$$

- Given a equilibrium point $f(x_0, u_0) = 0$
 recall that for $\dot{x} = f(x, u)$

$$\delta\dot{x} = \left. \frac{\partial f}{\partial x} \right|_{eq} \delta x + \left. \frac{\partial f}{\partial u} \right|_{eq} \delta u$$

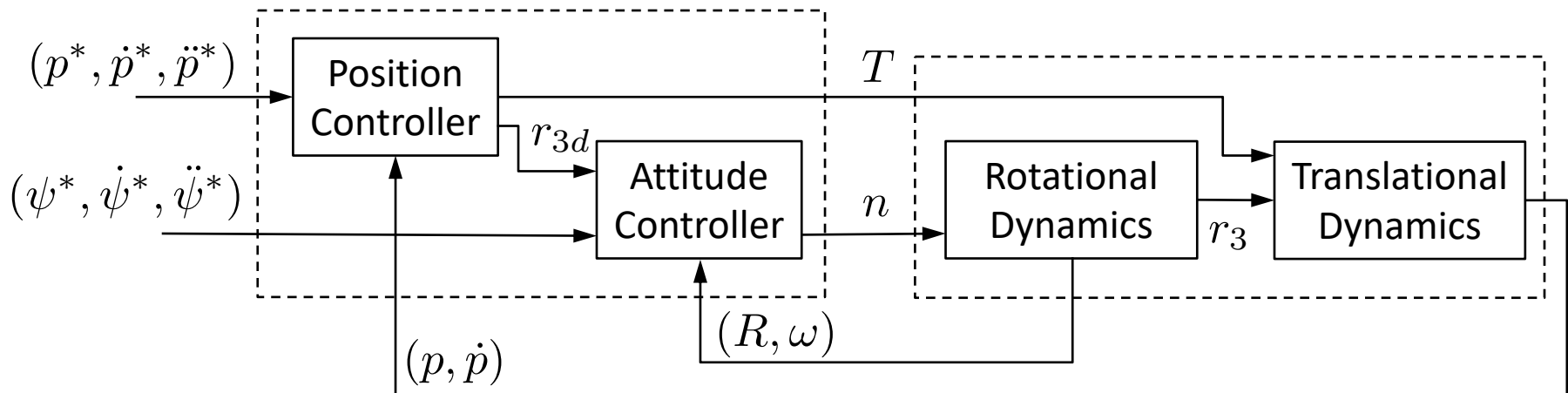
- Again a double integrator if J is diagonal

$$\delta\dot{\lambda} = \delta\omega$$

$$J\delta\dot{\omega} = \delta n$$

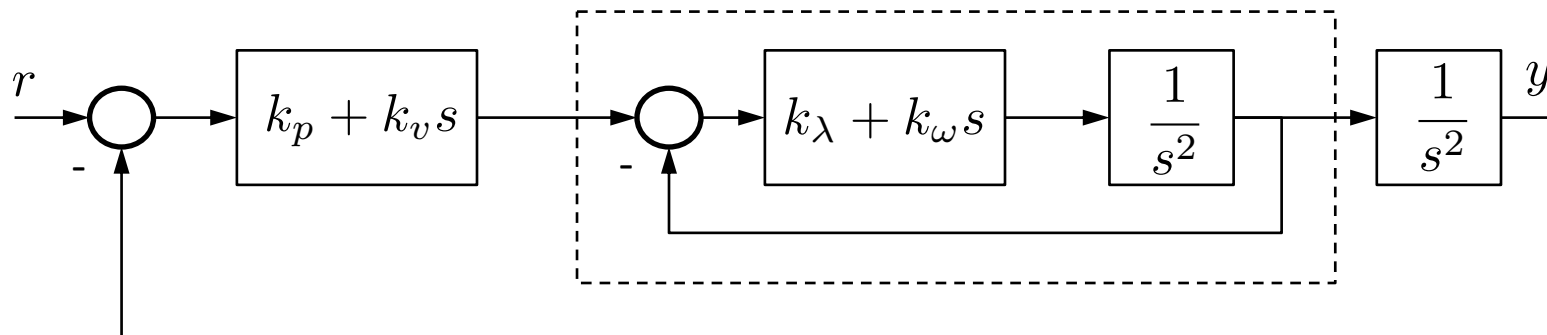
- PD controller $n = \delta n = -k_1(\lambda - \lambda_d) - k_2\omega$

Hierarchical control



- **Simplified SISO model**

- Bandwidth of inner-loop system 5 to 10 times higher than bandwidth of outer-loop system.

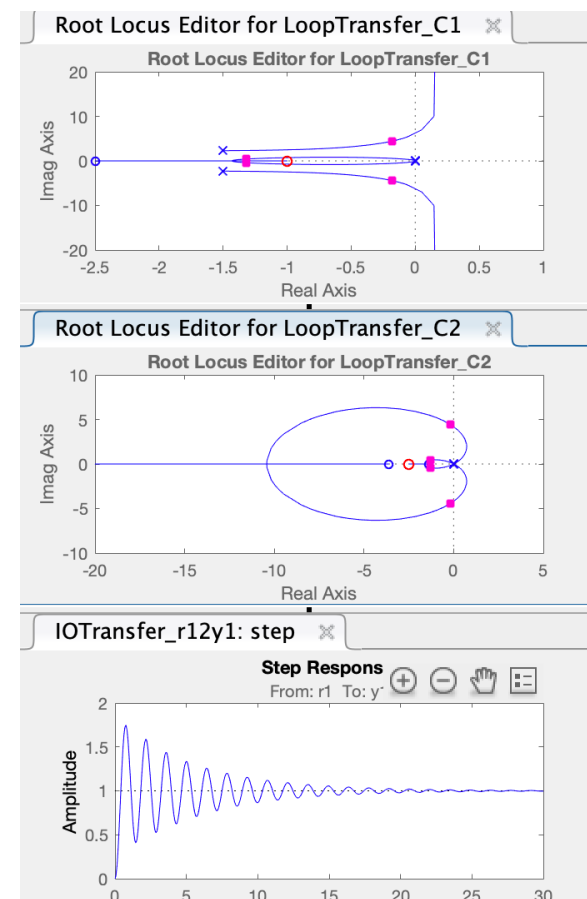
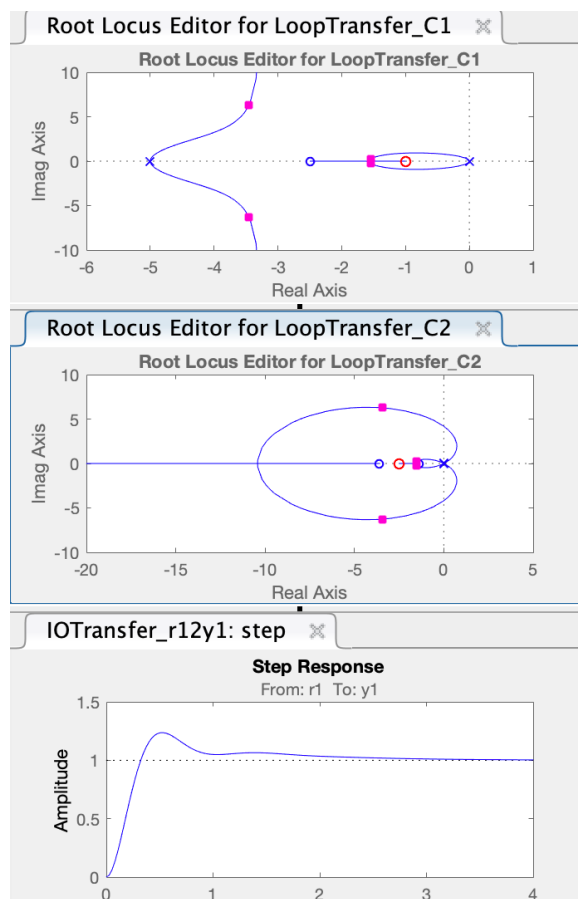


Hierarchical control

- Bandwidth of inner-loop system 5 to 10 times higher than bandwidth of outer-loop system.

$$C_1(s) = k_p + k_v s$$

$$C_2(s) = k_\lambda + k_\omega s$$



$$C_1(s) = 5(s + 1)$$

$$C_2(s) = 10(s + 2.5)$$

$$C_1(s) = 5(s + 1)$$

$$C_2(s) = 3(s + 2.5)$$

State-space representation – example

- Double integrator system

$$\ddot{y} = u$$

- State vector

$$x = \begin{bmatrix} y \\ \dot{y} \end{bmatrix}$$

- State space representation

$$\dot{x} = \underbrace{\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}}_A x + \underbrace{\begin{bmatrix} 0 \\ 1 \end{bmatrix}}_B u$$

- Solution of the unforced system ($u=0$)

$$x(t) = e^{At}x(0)$$

State-space representation - example

- Double integrator system
 - State space representation

$$\dot{x} = \underbrace{\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}}_A x + \underbrace{\begin{bmatrix} 0 \\ 1 \end{bmatrix}}_B u \quad x = \begin{bmatrix} y \\ \dot{y} \end{bmatrix}$$

- Solution of the unforced system ($u=0$)

$$\begin{aligned} x(t) &= e^{At} x(0) \\ &= \left(e^{\lambda_0 t} v_1 w_1^T + e^{\lambda_0 t} v_2 w_2^T + t e^{\lambda_0 t} v_1 w_2^T \right) x(0) \end{aligned}$$

$$\lambda_0 = 0 \quad v_1 = w_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad v_2 = w_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$x(t) = y(0)v_1 + \dot{y}(0)v_2 + \dot{y}(0)t v_1 = \begin{bmatrix} y(0) + \dot{y}(0)t \\ \dot{y}(0)t \end{bmatrix}$$

Position control – Pole placement

- Double integrator system – PID Control
 - State space representation of error system

$$x_e = \begin{bmatrix} y - r \\ \dot{y} - \dot{r} \end{bmatrix} \quad \dot{x}_e = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} x_e + \begin{bmatrix} 0 \\ 1 \end{bmatrix} (u - \ddot{r})$$

- PID controller

$$u = -k_P e - k_D \dot{e} - k_I \int e d\tau + \ddot{r}$$

- Augment the state with x_I

$$\dot{x}_I = e$$

$$x = \begin{bmatrix} x_I \\ x_e \end{bmatrix} \quad \dot{x} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} x + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} (u - \ddot{r})$$

$$u = - \underbrace{\begin{bmatrix} k_I & k_P & k_D \end{bmatrix}}_K x + \ddot{r}$$

Position Control – Pole placement

- Double integrator system – PID Control

- Closed-loop error system

$$\dot{x} = (A - BK)x = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -k_I & -k_P & -k_D \end{bmatrix} x$$

- Characteristic equation

$$\det(sI - (A - BK)) = s^3 + k_D s^2 + k_P s + k_I = 0$$

- Desired poles and desired characteristic equation

$$(s - p_1)(s - p_2)(s - p_3) = 0$$

$$s^3 - (p_1 + p_2 + p_3)s^2 + (p_1p_2 + p_1p_3 + p_2p_3)s - p_1p_2p_3 = 0$$

- Direct matching

$$k_D = -(p_1 + p_2 + p_3)$$

$$k_P = (p_1p_2 + p_1p_3 + p_2p_3)$$

$$k_I = -p_1p_2p_3$$

Position Control – Pole placement

- Double integrator system – PID Control

- Closed-loop error system

$$\dot{x} = (A - BK)x = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -k_I & -k_P & -k_D \end{bmatrix} x$$

- Direct matching for pole placement

$$k_D = -(p_1 + p_2 + p_3)$$

$$k_P = (p_1 p_2 + p_1 p_3 + p_2 p_3)$$

$$k_I = -p_1 p_2 p_3$$

- Examples

$$p_1 = -2; p_2 = -2; p_3 = 0$$

$$k_P = 4; k_D = 4; k_I = 0$$

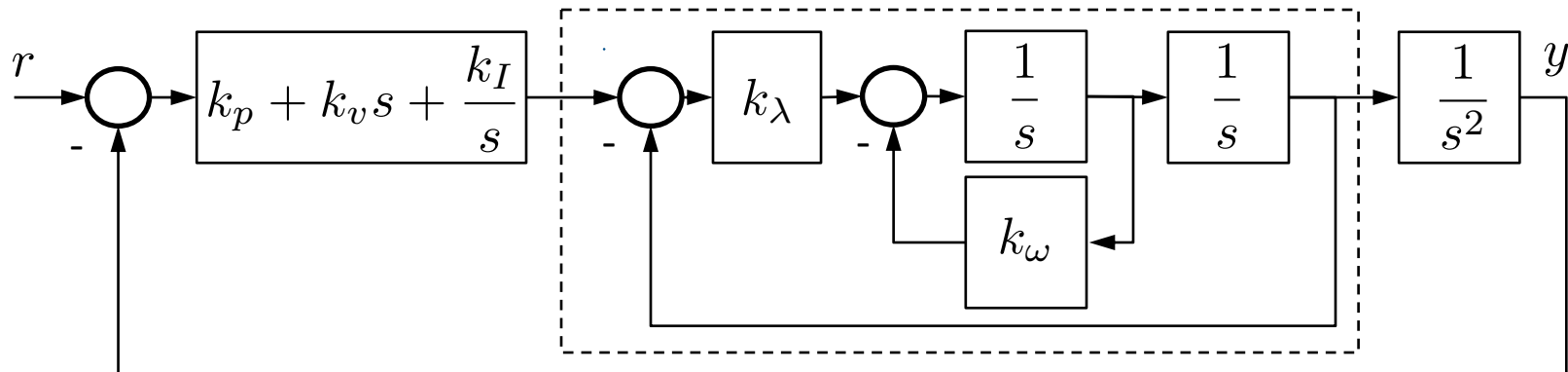
$$p_1 = -2; p_2 = -2; p_3 = -0.5$$

$$k_P = 6; k_D = 4.5; k_I = 2$$

- Recovers PD controller

Hierarchical Control – Pole placement

- Double integrator system + integrator + inner-loop dynamics



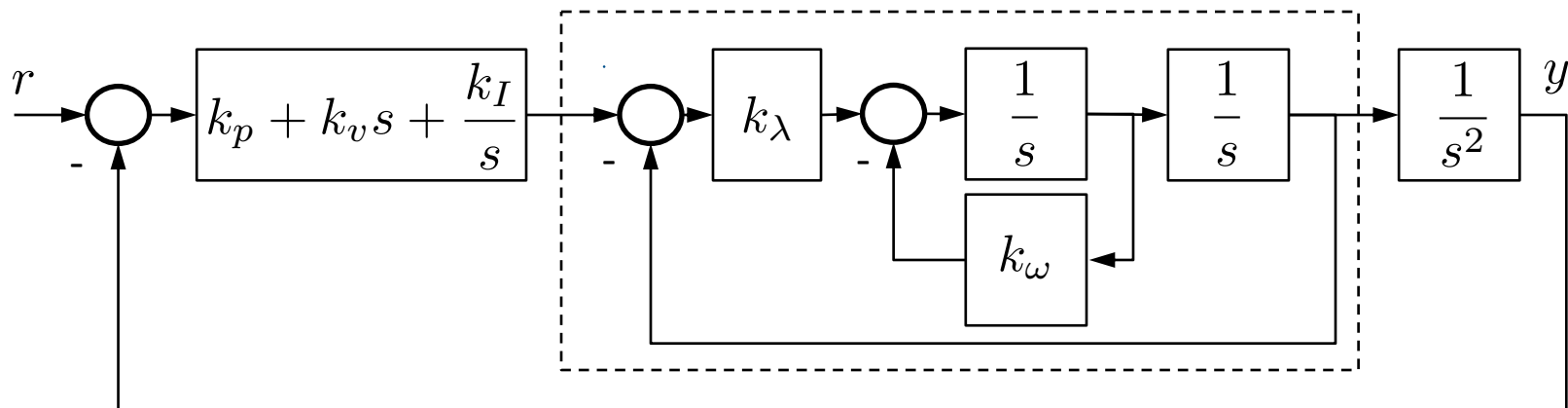
- Assume inner-loop system is fixed (k_λ, k_ω)
- State-space representation

$$\dot{x}_{in} = \begin{bmatrix} 0 & 1 \\ -k_\lambda & -k_\omega \end{bmatrix} x_{in} + \begin{bmatrix} 0 \\ k_\lambda \end{bmatrix} \theta_r \quad \dot{x} = Ax - B\theta, \quad \theta = \begin{bmatrix} 1 & 0 \end{bmatrix} x_{in}$$

$$\begin{bmatrix} \dot{x} \\ \dot{x}_{in} \end{bmatrix} = \begin{bmatrix} A & B \begin{bmatrix} 1 & 0 \end{bmatrix} \\ 0 & \begin{bmatrix} 0 & 1 \\ -k_\lambda & -k_\omega \end{bmatrix} \end{bmatrix} \begin{bmatrix} x \\ x_{in} \end{bmatrix} + \begin{bmatrix} 0 \\ \begin{bmatrix} 0 \\ k_\lambda \end{bmatrix} \end{bmatrix} \theta_r$$

Hierarchical Control – Pole placement

- Double integrator system + integrator + inner-loop dynamics



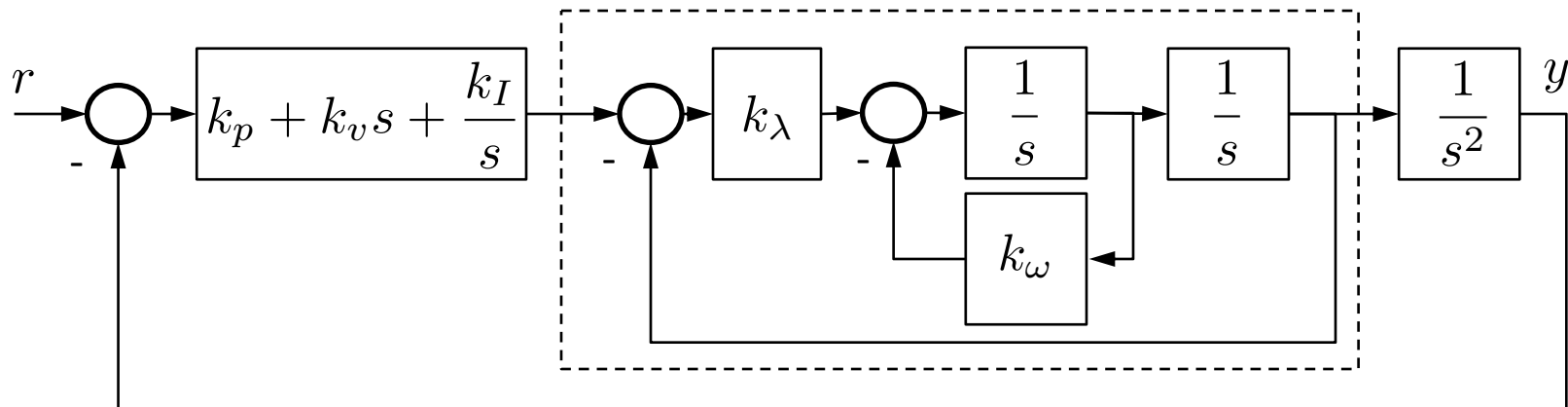
- State-space representation

$$\begin{bmatrix} \dot{x} \\ \dot{x}_{in} \end{bmatrix} = \begin{bmatrix} A & B \begin{bmatrix} 1 & 0 \end{bmatrix} \\ 0 & \begin{bmatrix} 0 & 1 \\ -k_\lambda & -k_\omega \end{bmatrix} \end{bmatrix} x_{in} + \begin{bmatrix} 0 \\ \begin{bmatrix} 0 \\ k_\lambda \end{bmatrix} \end{bmatrix} \theta_r$$

- Keep inner-loop poles (roots([1,kw,kl]))
- Place additional poles at -0.5+0.2i, -0.5-0.2i, -0.1

Pole placement example

- Double integrator system + integrator + inner-loop dynamics



- $K = \text{place}(A2, B2, [-0.5+0.2*i; -0.5-0.2*i; -0.1; \text{roots}([1, k_w, k_I])])$

$$u = \theta_r = \begin{bmatrix} 0.0290 & 0.4190 & 1.4973 & 1.1975 & 0.2750 \end{bmatrix} \begin{bmatrix} x_I \\ y - r \\ \dot{y} - \dot{r} \\ \theta \\ \dot{\theta} \end{bmatrix}$$

- No longer hierarchical!